

Principal Component Analysis

Unsupervised Learning - Representational Learning

Given a set of datapoints understand something useful about them !

Input

$$\{x_1, x_2, \dots, x_n\} \mid x \in \mathbb{R}^d$$

Output

Some Compressed representation of dataset

Example

$$\mathcal{D} = \left\{ \begin{bmatrix} -7 \\ -14 \end{bmatrix}, \begin{bmatrix} 2.5 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

Compress the dataset

for example using $\begin{bmatrix} 2.5 \\ 5 \end{bmatrix} \in \mathbb{R}^2$ as representative

$$v = c \times \begin{bmatrix} 2.5 \\ 5 \end{bmatrix} = \begin{bmatrix} 2.5c \\ 5c \end{bmatrix}$$

$$\therefore x = 2.5 \times c \implies c = \frac{x}{2.5} \quad (1)$$

$$\therefore y = 5 \times c \implies c = \frac{y}{5} \quad (2)$$

You can either use (1) with x coordinate or (2) with y coordinate to find the coefficient c

$$\begin{aligned}\begin{bmatrix} -7 \\ -14 \end{bmatrix} &\Rightarrow c = \frac{-14}{5} = \frac{-7}{2.5} = -2.8 \\ \begin{bmatrix} 2.5 \\ 5 \end{bmatrix} &\Rightarrow c = \frac{5}{5} = \frac{2.5}{2.5} = 1 \\ \begin{bmatrix} 1 \\ 2 \end{bmatrix} &\Rightarrow c = \frac{2}{5} = \frac{1}{2.5} = 0.4 \\ \begin{bmatrix} 0 \\ 0 \end{bmatrix} &\Rightarrow c = \frac{0}{5} = \frac{0}{2.5} = 0\end{aligned}$$

$\therefore$ Coefficients we get by using $\begin{bmatrix} 2.5 \\ 5 \end{bmatrix} \in \mathbb{R}^2$ as representative are $\{-2.8, 1, 0.4, 0\}$

Can we reconstruct the original dataset ?

representative $\times$ coefficients

$$\begin{bmatrix} 2.5 \\ 5 \end{bmatrix} \times \{-2.8, 1, 0.4, 0\} = \left\{ \begin{bmatrix} -7 \\ -14 \end{bmatrix}, \begin{bmatrix} 2.5 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$$

But using $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2$ as representative we cannot reconstruct the original dataset

Visualise the dataset

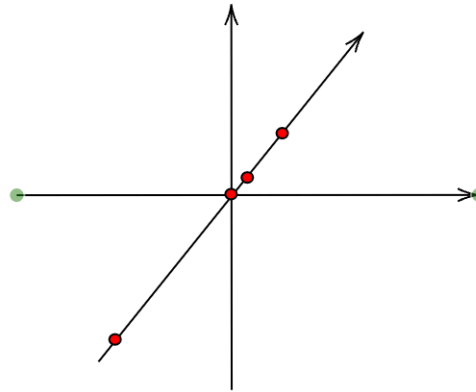


Figure 1: all the data points lying on a same line

If all the points lie on straight line, then using representative and coefficients we can reconstruct the original dataset, i.e, length of reconstruction error is zero

What have we gained compressing the dataset ?

Reduction in storage !

Real numbers needed to store original dataset = $d \times n$

Real numbers needed to store compressed dataset = $d + n$

What if all the data points doesn't lie on same line?

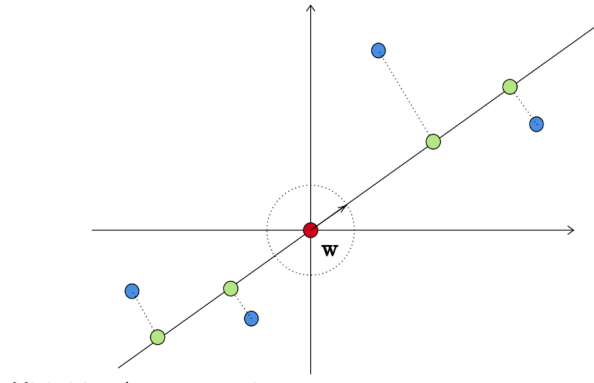


Figure 2: all the data points doesnot lie on same line

We have two options to consider

1. Give up exact reconstruction
2. Increase number of representative (but thats computationally inefficient)

So, we need to give up exact representation and find the proxy of the data point on to the line. Proxy should be choosen such that we loose least information - best case is projection of datapoint x onto representative w .

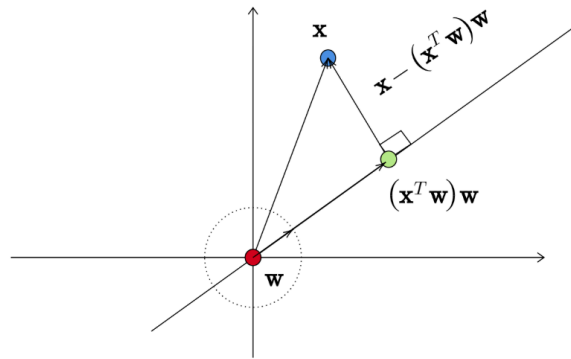


Figure 3: Labeled Diagram

- Datapoint = x
- Representative vector = w
- Error Vector = $x - (x^T w)w$ (assumption is that w is unit vector, $\|w\|^2 = 1$)
- Scalar(c) = $x^T w$ (assumption is that w is unit vector, $\|w\|^2 = 1$)
- Projection Vector = $(x^T w)w$ (assumption is that w is unit vector, $\|w\|^2 = 1$)

Objective Function

$$f(c) = \min_c \text{length}^2(\text{error vector})$$

To minimize:

$$f(c) = (x_1 - cw_1)^2 + (x_2 - cw_2)^2$$

Step 1: Expand f(c)

$$\begin{aligned}f(c) &= (x_1 - cw_1)^2 + (x_2 - cw_2)^2 \\&= x_1^2 - 2x_1cw_1 + c^2w_1^2 + x_2^2 - 2x_2cw_2 + c^2w_2^2 \\&= (x_1^2 + x_2^2) - 2c(x_1w_1 + x_2w_2) + c^2(w_1^2 + w_2^2).\end{aligned}$$

Step 2: Differentiate f(c) with respect to (c)

$$\frac{df}{dc} = -2(x_1w_1 + x_2w_2) + 2c(w_1^2 + w_2^2).$$

Step 3: Set $\frac{df}{dc} = 0$

$$\begin{aligned}-2(x_1w_1 + x_2w_2) + 2c(w_1^2 + w_2^2) &= 0 \\2c(w_1^2 + w_2^2) &= 2(x_1w_1 + x_2w_2) \\c &= \frac{x_1w_1 + x_2w_2}{w_1^2 + w_2^2}.\end{aligned}$$

Final Answer:

$$c = \frac{x_1w_1 + x_2w_2}{w_1^2 + w_2^2}$$

We can always pick $\begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ such that $\text{length} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = 1$

$$c^* = (x^T w)$$

$$\text{projection vector} = (x^T w) \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

There can be infinitely many representative lines but which line is the best representative ?

Given the dataset, find the line that has the least reconstruction error

Dataset

$$\{x_1, x_2, \dots, x_n\} \mid x \in \mathbb{R}^d$$

Error(line , dataset)

$$\begin{aligned} & \sum_{i=1}^n \text{error}(\text{line}, x_i) \\ & \sum_{i=1}^n \text{length}^2(x - (x^T w)w) \\ & \sum_{i=1}^n \|x_i - (x_i^T w)w\|^2 \\ & \frac{1}{n} \sum_{i=1}^n \|x_i - (x_i^T w)w\|^2 \\ & \frac{1}{n} \sum_{i=1}^n (x_i - (x_i^T w)w)^T (x_i - (x_i^T w)w) \\ & \frac{1}{n} \sum_{i=1}^n [x_i^T x_i - (x_i^T w)^2 - (x_i^T w)^2 + (x_i^T w)^2 w^T w] \\ & \frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2 \end{aligned}$$

$$g(w) = \frac{1}{n} \sum_{i=1}^n - (x_i^T w)^2 \left[\text{as } \frac{dg(w)}{dw} = 0 \right]$$

$$\min_{w : ||w||^2 = 1} g(w) = \frac{1}{n} \sum_{i=1}^n - (x_i^T w)^2$$

$$\max_{w : ||w||^2 = 1} g(w) = \frac{1}{n} \sum_{i=1}^n (x_i^T w)^2 \text{ [due to } g(w) \text{ being convex function]}$$

$$\frac{1}{n} \sum_{i=1}^n (w^T x_i) (x_i^T w)$$

$$\frac{1}{n} \sum_{i=1}^n w^T (x_i x_i^T) w$$

$$w^T \left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right) w$$

Equivalently, $\max_{w : ||w||^2 = 1} w^T C w$; where C is a covariance matrix of shape $d \times d$

w is the eigen vector corresponding to the maximum value of C

What does the best line represent?

The line that best represent the data in terms of the error minimisation of reconstruction is same as the line which maximizes the $w^T C w$ as C is the covariance matrix and this line is given by the eigen vector corresponding to the maximum eigen value of the covariance matrix.

Should we throw away the residue ?

$$\begin{array}{c} x \in \mathbb{R}^d \\ \downarrow \\ (x^T w) w \\ \downarrow \\ x - (x^T w) w \end{array}$$

This $x - (x^T w) w$ might not be error but contain some information

Dataset in Round 1

$$\begin{aligned} & \{x_1, x_2, \dots, x_n\} \mid x \in \mathbb{R}^d \\ & w_1 = \underset{w: ||w||^2 = 1}{argmax} w^T C w \\ & C = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right) \end{aligned}$$

$$\begin{aligned}
 x'_1 &\rightarrow x_1 - (x_1^T w_1) w_1 \\
 x'_2 &\rightarrow x_2 - (x_2^T w_1) w_1 \\
 &\vdots \\
 x'_n &\rightarrow x_n - (x_n^T w_1) w_1
 \end{aligned}$$

Dataset in Round 2

$$\begin{aligned}
 &\{x'_1, x'_2, \dots, x'_n\} \mid x \in \mathbb{R}^d \\
 &w_2 = \underset{w: \|w\|^2 = 1}{\operatorname{argmax}} w^T C' w \\
 &C' = \left(\frac{1}{n} \sum_{i=1}^n x'_i x'^T_i \right)
 \end{aligned}$$

- All residue are orthogonal to w_1
- The relationship between principal components are that they are perpendicular to each other
- Residue after d rounds: $\forall_i x_i = (x_i^T w_1) w_1 + (x_i^T w_2) w_2 + \dots + (x_i^T w_d) w_d$

What have we gained doing this?

If data points lie in lower dimension linear subspace then residue becomes zero much earlier than d-rounds

Example

Lets say our dataset is such that after 3 rounds residual becomes 0

Dataset

$$\{x_1, x_2, \dots, x_n\} \mid x \in \mathbb{R}^d$$

$$\forall_i x_i = (x_i^T w_1)w_1 + (x_i^T w_2)w_2 + (x_i^T w_3)w_3 \text{ such that } \{w_1, w_2, w_3\} \in \mathbb{R}^{100}$$

Representative

$$\{w_1, w_2, w_3\}$$

Coefficients

$$\{x_i^T w_1, x_i^T w_2, x_i^T w_3\}$$

Real numbers nedded to store original dataset = $d \times n$

Real numbers nedded to store compressed dataset = $d \times k + k + n$

here k is the number of round after which residue becomes zero !

What does eigen value of C mean ?

we know,

$$Cw_1 = \lambda_1 w_1$$

$$w_1^T C w_1 = w_1^T \lambda_1 w_1$$

$$\lambda_1 = w_1^T C w_1$$

$$\lambda_1 = w_1^T \left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right) w_1$$

$$\lambda_1 = \frac{1}{n} \left(\sum_{i=1}^n x_i^T w_1 \right)^2$$

How many direction is good enough?

$$\left(\frac{\sum_{i=1}^l \lambda_i(C)}{\sum_{i=1}^d \lambda_i(C)} \right) \geq 0.95$$

$$\frac{\lambda_1 + \dots + \lambda_k}{\lambda_1 + \dots + \lambda_d} \geq 0.95$$

Dataset Centering

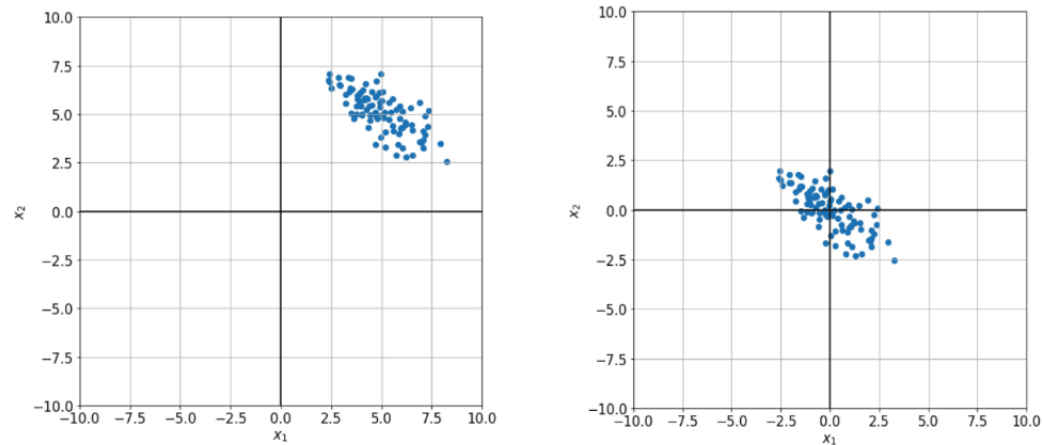


Figure 4: Dataset Centering

Data centering is essential in PCA because it ensures that each feature has zero mean, allowing the covariance matrix to accurately capture the true directions of variance. Without centering, the principal components may be biased by the mean, leading to incorrect or misleading results. Centering aligns PCA with the geometric structure of the data by focusing on variation around the data's center rather than the origin.

What does this proxies mean ?

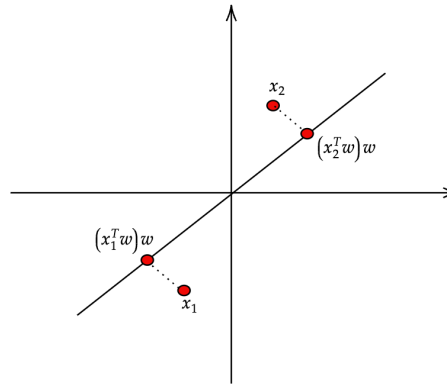


Figure 5: datapoint showing proxies on representative line

$$\left\{ \left(x_1^T w \right) , \dots , \left(x_n^T w \right) \right\}$$

Average of Proxies

$$\frac{1}{n} \left(\sum_{i=1}^n x_i^T w \right) = \left(\frac{1}{n} \sum_{i=1}^n x_i \right)^T w = 0^T w = 0$$

Variance of Proxies

$$\frac{1}{n} \left(\sum_{i=1}^n x_i^T w - \text{mean} \right)^2 = \frac{1}{n} \sum_{i=1}^n \left(x_i^T w \right)^2$$

To Conclude, error minimisation on centered dataset $\equiv$ Variance maximisation

Why do we study PCA ?

In general, we want to find directions, that has the highest variance and the direction is known as principal component $\{w_1, w_2, \dots, w_d\}$

Steps Involved in PCA

Step 1: Center the dataset

$$X_c = X - X_{mean}$$

Step 2: Calculate the covariance matrix of the centered data

$$C = \left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right)$$

Step 3: Compute the eigenvectors and eigenvalues – [watch here](#)

Step 4: Sort the eigenvalues in descending order and choose the top k eigenvectors corresponding to the highest eigenvalues

Step 5: Transform the original data by multiplying it with the selected eigenvectors(PC's) to obtain a lower-dimensional representation.

Reference

Arun Raj Kumar, Lecture Slides for Machine Learning Techniques

Karthik Thiagarajan, Machine Learning Techniques Github Notes.

Indian Institute of Technology Madras , CS2007 Course Website

Stanford University, CS229 Machine Learning Notes

Thank You !